WORK REPORT

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Simulation Result of Heptane molecule(UCCSD)

Simulation Specifications:

- maxiter= 2000, maxfunc = 10000
- Active Space = 6e6o
- Ansatz = UCCSD

Similar Simulation is done for other 11 molecules i.e Propane to Nonane And Benzene to Pentacene.

Func.	DFT	Embedd	VQE	Lib Import	Ansatz Build	Setup & Int	Classical Qiskit Ovh.	Total Prof.	Real Time	Unprof. Gap
lda_rs	30.55	16.27	**2594.61**	28.29	0.27	7.48	327.59	3005.06	3009.08	**4.02s**
camb3lyp	51.53	41.90	**7880.16**	42.13	0.42	7.78	822.27	8846.19	8857.70	**11.51s**

VQE TIME SPLIT Details (UCCSD Ansatz)

Functional	Embed Iters	VQE Iters	Energy Eval (s)	Gradient Eval (s)	Optimizer Time (s)	Total Iter Time (s)	Energy Convergence
lda_rs	1	590	1.986	0.000	1289.247	1291.233	0
	2	590	2.082	0.005	1301.285	1303.372	7.17E-10
	Total (2)	**1180**	**4.068**	**0.005**	**2590.532**	**2594.605**	
camb3lyp	1	708	2.375	0.000	1559.484	1561.859	0
	2	708	2.401	0.004	1575.320	1577.725	1.15E-07
	3	708	2.781	0.005	1566.469	1569.255	1.58E-08
	4	708	2.368	0.006	1560.073	1562.447	1.64E-09
	5	708	2.563	0.004	1606.294	1608.861	7.89E-09
	Total (5)	**3540**	**12.488**	**0.019**	**7867.640**	**7880.147**	

Simulation Result of Heptane molecule(IITB)

Simulation Specifications:

- maxiter= 2000, maxfunc = 10000
- Active Space = 6e6o
- Ansatz = IITB

Similar Simulation is done for other 11 molecules i.e Propane to Nonane And Benzene to Pentacene.

Func.	DFT	Embedd	VQE	Lib Import	Ansatz Build	Setup & Int	Classical Qiskit Ovh.	Total Prof.	Real Time	Unprof. Gap
lda_rs	29.02	16.47	**37.75**	39.99	66.31	7.17	36.97	233.68	238.38	**4.70s**
camb3lyp	46.51	41.42	**127.98**	44.58	69.76	7.23	93.26	430.74	435.44	**4.70s**

VQE TIME SPLIT Details (IITB Ansatz)

Functional	Embed Iters	VQE Iters	Energy Eval (s)	Gradient Eval (s)	Optimizer Time (s)	Total Iter Time (s)	Energy Convergence
lda_rs	1	75	0.092	0.000	18.844	18.936	0
	2	75	0.095	0.002	18.713	18.810	4.71E-11
	Total (2)	**150**	**0.187**	**0.002**	**37.557**	**37.746**	
camb3lyp	1	100	0.122	0.000	25.044	25.166	0
	2	100	0.134	0.003	25.336	25.473	1.15E-07
	3	100	0.134	0.003	25.563	25.700	2.16E-08
	4	100	0.132	0.003	25.474	25.609	6.69E-09
	5	100	0.133	0.003	25.888	26.024	2.68E-09
	Total (5)	**500**	**0.655**	**0.012**	**127.305**	**127.972**	

OTHER COMPLETIONS:

- Contributed a section in chapter 6 of the HPCQC Booklet.
- Updated the booklet section with new results,mentioning the insights.
- Contributed in making of Github Repo with Tejjan, containing detailed README’s for future use.

THANK YOU